Name:

Collaborators:

Instructions: Worksheets are graded mostly on completion and partially on correctness. Please write complete solutions showing explanations and key steps to the following problems, unless it says otherwise.

Applying Lagranges Method for Linear Systems

1. The Wronskian for Linear Systems

Consider this *non-homogeneous linear system of 1st-order ODEs with constant coefficients*:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \vec{f}(t).$$

Note the non-homogeneous term $\vec{f}(t)$, a vector function of t, added to the system. Let $\vec{f}(t) = \begin{bmatrix} 3e^{2t} \\ -e^{2t} \end{bmatrix}$.

The following tasks outlines the method of *Variation of Parameters* to find the particular solution of the linear system.

Task	Show work here
The eigenvalues and eigenvectors of the coefficient matrix are $\lambda_1 = 3$ with $\vec{v}_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ and $\lambda_2 = -1$ with $\vec{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Construct the matrix $X = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ where a and c are the elements of $\vec{v}_1 e^{\lambda_1 t}$, and c and d are the elements of $\vec{v}_2 e^{\lambda_2 t}$.	
—-	——–
Determine $W = \det(X)$.	
—-	——–
Determine $X^{-1} = \dfrac{1}{W} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.	
—-	——–
Determine the particular solution $\begin{bmatrix} x_p \\ y_p \end{bmatrix} = X \int X^{-1} \vec{f}(t)\, dt$.	

The matrix W is the Wronskian and the result of the integral $X \int X^{-1} \vec{f}(t)\, dt$ is the particular solution as long as $\vec{f}(t)$ is bounded and continuous. Note that the matrix X can be a constant or a function of t.

2. An Unusual Vector Function for the Non-Homogeneous Term

The method of Variation of Parameters is useful when the non-homogeneous term of the linear system contains a function that is not in the table of common functions for the method od undetermined coefficients.

Consider this *non-homogeneous linear system of 1st-order ODEs with constant coefficients*:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \vec{f}(t).$$

Note the non-homogeneous term $\vec{f}(t)$, a vector function of t, added to the system. Let $\vec{f}(t) = \begin{bmatrix} \ln(t) \\ e^{-t} \end{bmatrix}$. Because of the superposition principle, the full solution is of the form $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x_h \\ y_h \end{bmatrix} + \begin{bmatrix} x_p \\ y_p \end{bmatrix}$ where $\begin{bmatrix} x_h \\ y_h \end{bmatrix}$ is the homogeneous solution and $\begin{bmatrix} x_p \\ y_p \end{bmatrix}$ is the particular solution.

a. Determine the homogeneous solution $\begin{bmatrix} x_h \\ y_h \end{bmatrix}$ using the method of eigenvalues and eigenvectors.

b. Use Part (a) to construct the matrix X.

c. Determine the particular solution

$$\begin{bmatrix} x_p \\ y_p \end{bmatrix} = X \int X^{-1} \vec{f}(t)\ dt.$$

d. Write the general solution.